Education Experience:

I hold a BSc degree in Medical Physics with Physics from University College London. During my undergraduate studies, I focused on the application of electromyography (EMG) in biofeedback for my final project. This experience marked the beginning of my exploration into the complexities of the neuromuscular system and physiological signal. I used MATLAB for signal analysis tasks and translated the recorded arm EMG signal into sound in real-time with microcontroller. This creative project not only enhanced my problem-solving abilities but also deepened my interest in discovering the fundamental of the physiological representations.

I have recently completed a MSc degree in Biomedical Engineering with a specialization in the Neurotechnology stream at Imperial College London. This course equipped me with structured knowledge and research skills in the field of Neuroscience. I have gained a solid foundation in basic neuroscience principles and have learned about various methodologies for recording, processing, and analyzing neural data. In terms of programming, I have utilized both MATLAB and Python for tasks such as decoding neural time series data and modeling neural networks.

For my individual MSc project, I attempted to monitor calcium dynamics in brain organoids derived from induced pluripotent stem cells. This in vitro research provided me with experience in tissue culture for sample maintenance and microscopy for data collection. The combination of my background knowledge, laboratory experience, and specialized training are transferable to this applied work.

Additional Laboratory Experience:

During the summer internship, I had the opportunity to work in a biomedical ultrasound laboratory. My main task was to design experiments aimed at evaluating the performance of a launched ultrasound therapy system. This work differed from my previous experiences of being supervised on undergraduate and master’s projects, as it demanded my decision-making skills and responsibility for the research progress. Over the course of two months, I gained valuable practical experience that would be beneficial for both my future academic studies and research applications.

Motivation and Future Plan:

This opportunity to delve into the fundamental mechanisms of the nervous system function for the purpose of developing treatments aligned with my interests. The previous research experiences left me with lots of curiosity and many unanswered questions from the basic principles in neuroscience to the actual functional outcomes. I believe that this opportunity could provide me with valuable knowledge and in-depth experience for applying a PhD position and carrying research in the future.